

LUCKY ADOH

Research Engineer

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PROFESSIONAL SUMMARY

Research engineer specialising in mathematical and computational modelling, optimisation, structural dynamics, and transport engineering. Experienced in developing and validating engineering models, numerical simulation, sensitivity analysis, experimental testing, and data analysis. Combines modelling and experimental methods to investigate complex engineering systems, with research interests in optimisation, transport and energy systems, infrastructure decision support, and sustainable engineering.

WORK EXPERIENCE

PhD Researcher in Vibration and Structural Dynamics

University of Southampton, UK | May 2023 to Sept 2026

- ▶ Conducted experimental and numerical analysis of structural vibration under transient dynamic loading.
- ▶ Performed vibration testing, instrumentation, data acquisition, and signal processing using FFT, PSD, and frequency-domain analysis.
- ▶ Developed and validated numerical models of structural and electromechanical dynamic systems using MATLAB and Python.
- ▶ Investigated resonance, damping, nonlinear dynamics, and vibration control through experimental and computational methods.
- ▶ Designed and experimentally validated vibration-based sensing and energy harvesting systems for structural and railway applications.
- ▶ Conducted field and laboratory vibration measurements and validated computational predictions against experimental data.

Research Officer

*Institute of Materials Research and Engineering, A*STAR, Singapore* | March 2025 to Sept 2025

- ▶ Conducted experimental and numerical analysis of piezoelectric transducers for vibration-based energy harvesting applications.
- ▶ Designed and tested nonlinear energy harvesting mechanisms to increase power output from broadband vibration sources.
- ▶ Integrated sensor systems for real-time structural health monitoring, supporting continuous infrastructure integrity assessment.
- ▶ Collaborated with interdisciplinary teams applying experimental methods and statistical analysis to support research outcomes.
- ▶ Supported experimental design, data interpretation, and validation for ongoing research projects.

Demonstrator / Postgraduate Teaching Assistant

University of Southampton, UK | Oct 2023 to Sept 2026

- ▶ Delivered laboratory support, guiding students through experimental procedures and technical concepts in structural dynamics and sensor systems.
- ▶ Marked and provided constructive feedback on lab reports and logbooks.
- ▶ Taught tutorials and supported students in understanding engineering principles, data interpretation, and problem-solving techniques

PhD Tutor

The Brilliant Club, UK | Oct 2023 to Dec 2025

- ▶ Supported and inspired young people (KS3 to KS5) from less advantaged backgrounds to access university education.
- ▶ Communicated advanced engineering research to non-specialist audiences.

EDUCATION

PhD in Engineering and the Environment

University of Southampton, UK | May 2023 to Sept 2026 (Expected)

MSc in Railway Engineering

Addis Ababa University

BEng in Mechanical Engineering

Federal University of Technology, Akure

SKILLS AND EXPERTISE

Computational Modelling and Optimisation

- ▶ Mathematical and numerical modelling of mechanical, structural, and coupled electromechanical systems.
- ▶ Parameter optimisation, sensitivity analysis, design-space exploration, and evaluation of engineering performance under different operating conditions.
- ▶ MATLAB and working knowledge of Python for numerical simulation, data analysis, signal processing, and visualisation.
- ▶ Finite element analysis, model validation, and comparison of numerical predictions with experimental measurements.

Transport and Infrastructure Engineering

- ▶ Railway dynamics and infrastructure, including train-induced vibration, switches and crossings, and track condition assessment.
- ▶ Analysis of infrastructure condition data, deterioration, and maintenance prioritisation.
- ▶ Engineering analysis to support infrastructure monitoring, maintenance, and operational decision-making.

Structural Dynamics and Experimental Engineering

- ▶ Structural dynamics, modal analysis, FFT, PSD, frequency-response analysis, resonance, damping, and transient response.
- ▶ Experimental vibration testing, instrumentation, sensors, data acquisition, and dynamic characterisation.
- ▶ Design and execution of laboratory and field experiments for validation of engineering models.

Energy and Electromechanical Systems

- ▶ Modelling and experimental characterisation of piezoelectric and electromagnetic energy harvesting systems.
- ▶ Linear, nonlinear, and piecewise-linear systems under harmonic, transient, and impulsive excitation.
- ▶ Electromechanical energy conversion, power optimisation, energy storage, and self-powered sensing.

PATENTS, PUBLICATIONS & CONFERENCES

Patent application

[P1] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, "*Multiple harvester system for impulsive vibration at railway track discontinuities*," U.K. Patent Application No. 2606823.9, filed Mar. 27, 2026.

Selected peer review publications

- [J1] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, K. Yao, Y.M. Yousry, V.K. Wong, "*Transient Impulsive energy harvesting from Railway Switches and Crossings*," Railway engineering Science, 2026, DOI: [10.1007/s40534-026-00459-y](https://doi.org/10.1007/s40534-026-00459-y)
- [J2] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, S. C. Lai, K. Yao, "*Comparative experimental assessment of d_{31} and d_{33} piezoelectric harvesters for railway energy harvesting*," 2026, DOI: [10.1109/JSEN.2026.3731419](https://doi.org/10.1109/JSEN.2026.3731419)
- [J3] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, M. Toward, "*Experimental comparative study of energy harvesting systems for Railway Switches and Crossings*," 2026, **Under Review**.
- [J4] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, "*Piecewise linear electromagnetic energy harvester for transient railway vibration*," 2026, **Under Review**.
- [J5] L. Adoh, L. Adoh, R. M. Getahun, D. Thompson, T. Mencha, B. Gashaw, K. Tesfaye, "*Spatiotemporal Analysis of Railway Track Geometry Defects from Repeated Dynamic Inspections: Evidence from the Ethio–Djibouti Railway*," 2026, **Under Review**.
- [J6] L. Adoh, D. Thompson, D. Yurchenko, and G. Squicciarini, 'Performance of a Piezoelectric Energy Harvester under Harmonic and Transient Railway Excitation', in Proc. of the 14th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems, *ENSsys*, Saint Malo, France, 2026. DOI: [10.1109/ENSsys71150.2026.00014](https://doi.org/10.1109/ENSsys71150.2026.00014)
- [J7] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, "*Power harvesting from train-induced vibrations at railway switches and crossings*," in Proc. of the 22nd International Conference on Transport Science, Transportation Research Procedia, 2026, in press.
- [J8] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, "*Modelling and Experimental Validation of Nonlinear Energy Harvesters for Railway Applications*," in Proc. 6th Conf. Structural Nonlinear Dynamics and Diagnosis, Marrakech, Morocco, 2026, in press.
- [J9] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, S.C. Lai, K. Yao "*Piezoelectric Transducers and their Energy Harvesting Applications: a Case Study of Railway Crossing*," in Proc. of the IEEE International Symposium on Applications of Ferroelectrics, Montreal, Canada, 2026, in press.
- [J10] L. Adoh, D. Yurchenko, D. Thompson, and G. Squicciarini, "*Influence of railway crossing geometry on vibration energy harvesting for self-powered railway monitoring systems*," in Proc. of 9th IEEE International Energy Conference, Paris, France, 2026, in press
- [J11] L. U. Adoh and S. Tesfaye, "*Study of the wear behaviour of rail material*," Int. J. Res. Appl. Eng., vol. 8(2), pp. 33–45, 2021, DOI: [10.22068/ijrare.287](https://doi.org/10.22068/ijrare.287)
- [J12] F. M. Akello and L. U. Adoh, "*Simulation of power generation from railway vibration*," Int. J. Sustainable Green Energy, vol. 9, pp. 16–22, 2020, DOI: [10.11648/j.ijrse.20200901.12](https://doi.org/10.11648/j.ijrse.20200901.12)
- [J13] Adoh, L.U, Akello F.M, Nyangassa F, Ishimwe P, "Prevention of Railway Accident using Arduino Based Safety System: A case Study of Addis Ababa Light Rail Transit". Intl Journal of Engr Res & Tech. 8(9): 327-332, 2019, DOI: [10.17577/IJERTV8IS090091](https://doi.org/10.17577/IJERTV8IS090091)
- [J14] Adoh L. U, Mutswatiwa L, Akello F.M. "Safety demonstration and risk management at rail-road level crossing at Addis Ababa light rail transit network". Intl Journal of Sci Res in Sci, Engr & Tech. 6(5): 103-109, 2019.

Conference presentations

- [C1] L. Adoh, 'Performance of a Piezoelectric Energy Harvester under Harmonic and Transient Railway Excitation', *14th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems, ENSys*, Saint Malo, France, 2026.
- [C2] L. Adoh, "Power harvesting from train-induced vibrations at railway switches and crossings," *22nd International Conference on Transport Science*, Portoroz, Slovenia, 2026.
- [C3] L. Adoh, "Modelling and Experimental Validation of Nonlinear Energy Harvesters for Railway Applications," *6th Conf. Structural Nonlinear Dynamics and Diagnosis*, Marrakech, Morocco, 2026.
- [C4] L. Adoh, "Piezoelectric Transducers and their Energy Harvesting Applications: A Case Study of Railway Crossing," *IEEE International Symposium on Applications of Ferroelectrics*, Montreal, Canada, 2026.
- [C5] L. Adoh, "Influence of railway crossing geometry on vibration energy harvesting for self-powered railway monitoring systems," *IEEE International Energy Conference*, Paris, France, 2026.
- [C6] L. Adoh, "Prediction of Train-induced Impulsive Vibration at Railway Crossing", *1st UKRRIN Student Conference*, Sheffield, United Kingdom, 2025.
- [C7] L. Adoh, "A Model for Train-induced Vibration at Railway Crossings for Energy Harvesting Applications", *14th International Conference on Recent Advances in Structural Dynamics*, Southampton, United Kingdom, 2024.

PROFESSIONAL MEMBERSHIPS

- Associate Fellow, Higher Education Academy, UK (Ref: PR334157)
- Corporate Member, Nigerian Society of Engineers (#65917)
- Chartered Engineer, COREN, Nigeria(#R78872)
- Graduate Member, Railway Engineering Institution (RailEI, UK) (#23-09909)
- Member, International Association of Engineers (IAENG #257005)

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

- Making a Point: Developments in Switches and Crossings, Railway Engineering Institution Technical Seminar, University of Manchester, March 2026.
- PRINCE2, IT Career Switch, 2026.
- Six Sigma Green Belt, IT Career Switch, 2026.
- Project Management Professional, IT Career Switch, 2026.
- Agile Project Management, IT Career Switch, 2026.
- Business Analysis Foundation, IT Career Switch, 2026.

GRANTS AND FUNDING

- Petroleum Technology Development Fund, (~£200,000)
- World bank African Railway Centre of Excellence MSc grant (~£35,000)
- Research Training Support Grant, University of Southampton (~£3,500)
- Research Training Support Grant, Dynamic group, ISVR, University of Southampton (~£500)
- IEEE-PFM Student research travel grant (~£400)

COMMUNITY ENGAGEMENT

PhD Mentor

University of Southampton | 2024 to 2026

- Provided guidance and support to new doctoral students navigating research challenges.

Volunteer

Railway Engineering Institution (RailEI), UK | July 2024

- Supported open day organisation and engineering outreach activities.

University of Southampton, Southampton, UK | 16 to 20 Sept 2026

- Help in organising the British Science Festival